

MEDLEDGER: A BLOCKCHAIN-BASED ANTI-COUNTERFEIT PHARMACEUTICAL SUPPLY CHAIN

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ER MODEL - ENTITIES AND ATTRIBUTES

ENTITY	PRIMARY KEY	OTHER ATTRIBUTES
PARTY	party_id (PK)	party_name, address, phone, email
MANUFACTURER (SUBCLASS)	party_id (PK, FK)	manufacturing_license_no
DISTRIBUTOR (SUBCLASS)	party_id (PK, FK)	distributor_license_no
PHARMACY (SUBCLASS)	party_id (PK, FK)	pharmacy_license_no, ho
REGULATOR (SUBCLASS)	party_id (PK, FK)	regulator_code
DRUG	drug_id (PK)	drug_name, description, strength, dosage_form
BATCH	batch_id (PK)	batch_status, manufacture_d
QUALITY_TEST	test_id (PK)	test_date, test_type, result status
RECALL	recall_id (PK)	recall_date, status, reason
PACKAGE	package_id (PK)	package_size, packaged_at, qr_code, status
SHIPMENT	shipment_id (PK)	shipment_date, status, mo
DISPENSING	dispense_id (PK)	dispensed_at, patient_id, quantity, remarks

FER Advanced Concepts

1) Generalization / Specialization

- PARTY is generalized into four subclasses - MANUFACTURER, DISTRIBUTOR, PHARMACY and REGULATOR connected through an ISA triangle marked disjoint, total in the diagram. This is attribute-defined, disjoint, total specialization:
 - Disjoint - every party plays exactly one role; a manufacturer cannot simultaneously be a pharmacy in this model.
 - Total (mandatory) - every PARTY row must belong to one of the four subclasses; there is no 'generic' party with no role.
 - Each subclass inherits all of PARTY's attributes (party_id, party_name, address, phone, email) and adds role-specific attributes (e.g. manufacturer_license_no, pharmacy_license_no).
- Mapping Choice: because the specialization is total and disjoint but the subclasses carry genuinely different attributes, the standard 'Rule 8b' mapping is used - one table for the superclass PARTY plus one table per subclass, each sharing party_id as both its PK and a FK referencing PARTY. This avoids the wide, mostly NULL single table that a 'single table with role attributes' mapping would produce, and keeps each subclass's optional attributes out of the other roles' rows.

2) Aggregation

- A plain ER relationship cannot participate in another relationship, but MEDLEDGER needs exactly that: the custody transfer event formed by PARTY - SENDS → SHIPMENT ← RECEIVES → PARTY is not just an operational record in Oracle - the whole SHIPMENT-with-its-SENDS/RECEIVES relationship is treated as a single higher-level object (an aggregation) that is subsequently mirrored as one custody-transfer transaction on the Hyperledger Fabric ledger, and again as one edge between two PARTY nodes in the Neo4j provenance graph. Aggregation is the correct FER construct here because the 'thing' being

replicated across layers is the relationship instance itself not a plain attribute of an entity.

A smaller, smaller aggregation is BATCH treated together with its ~~PACKED~~ PACKAGED-As relationship to PACKAGE: this combined 'traceable unit' (batch + the specific package) is what actually gets scanned, shipped and dispensed - i.e., downstream relationship attach to the batch-package pairing as a whole, not to BATCH or PACKAGE in isolation.

FUNCTIONAL DEPENDENCIES

- 1) PARTY: party-id $\rightarrow$ party-name, address, phone, email
- 2) MANUFACTURER: party-id $\rightarrow$ manufacturing-license-no
- 3) DISTRIBUTOR: party-id $\rightarrow$ ^{distributor} ~~pharmacy~~ license-no, ~~hq~~
- 4) PHARMACY: party-id $\rightarrow$ pharmacy-license, hq
- 5) REGULATOR: party-id $\rightarrow$ regulator-code
- 6) DRUG: drug-id $\rightarrow$ drug-name, description, strength, dosage-form
- 7) BATCH: batch-id $\rightarrow$ drug-id, manufacturer-id, recall-id, batch-status, manufacture-date
- 8) QUALITY-TEST: test-id $\rightarrow$ batch-id, test-date, result, status
- 9) RECALL: recall-id $\rightarrow$ recall-date, status, reason
- 10) PACKAGE: package-id $\rightarrow$ batch-id, dispense-id, package-size, packaged-at, qrs-code, status, quantity-total;
qrs-code $\rightarrow$ package-id, batch-id, dispense-id, package-size, ~~size~~ status, package-date, quantity-total
- 11) SHIPMENT: shipment-id $\rightarrow$ sender-party-id, receiver-party-id, status, shipment-date, mode
- 12) DISPENSING: dispense-id $\rightarrow$ pharmacy-id, dispensed-at, patient-id, quantity, remarks.

Mapping EER Model to a Relational Schema

Relation	Attributes	Foreign Keys
PARTY	party-id, party-name, address, phone, email	—
MANUFACTURER	party-id, manufacturing-licence-no	party-id → PARTY. party-id
DISTRIBUTOR	party-id, distributor-license-no	party-id → PARTY. party-id
PHARMACY	party-id, pharmacy-licence-no, hq	party-id → PARTY. party-id
REGULATOR	party-id, regulator-code	party-id → PARTY. party-id
DRUG	drug-id, drug-name, description, strength, dosage-form	—
BATCH	batch-id, drug-id, manufacturer-id, recall-id, batch-status, manufacture date	drug-id → DRUG.drug-id, manufacturer-id → M.party-id, recall-id → RECALL.recall-id
QUALITY-TEST	testid, drug-id, manufacturer-id, recall-id, batch-id, test-date, test-type, result, status	batch-id → BATCH. batch-id

Relation	Attributes	Foreign Keys
RECALL	recall_id, recall_date, status, reason	—
PACKAGE	package_id, batch_id, dispense_id, package_size, packaged_at, qr_code, status, quantity_total	batch_id → BATCH. batch_id, dispense_id → DISPENSING. dispense_id
SHIPMENT	shipment_id, sender_party_id, receiver_party_id, shipment_date, status, mode	sender_party_id → PARTY.party_id, receiver_party_id → PARTY.party_id
CONTAINS	package_id, shipment_id	package_id → PACKAGE.package_id, shipment_id → SHIP.shipment_id
DISPENSING	dispense_id, pharmacy_id, dispensed_at, patient_id, quantity, remarks	pharmacy_id → PHARMACY.party_id

Normalisation

Assume that the relations are initially in 1NF form.

- i) PARTY(party_id, party_name, address, phone, email),
manufacturing_license_no, distributor_license_no,
pharmacy_license_no, regulator_code)
- ii) BATCH(batch_id, manufacture_date, batch_status, drug_id,
drug_name, strength, dosage_form, manufacturer_id,
manufacturing_license_no, test_id, test_date,
test_type, result), recall_id, recall_date, status, reason)
- iii) PACKAGE(package_id, batch_id, dispense_id, package_size,
packaged_at, qr_code, status, quantity, pharmacy_id,
dispensed_at, patient_id, quantity, remarks,
shipment_id, sender_id, shipment_date, status)

Convert to 2NF

- i) Party is already in 2NF
- ii) Splitting the table into the attributes dependent on
batch_id, test_id, ~~and recall_id~~,

BATCH(batch_id, manufacture_date, batch_status,
drug_id, drug_name, strength, dosage_form,
manufacturer_id, manufacturing_license_no,
recall_id, recall_date, status, reason)

TEST(test_id, test_date, test_type, result)

iii) Splitting the relation into ^{attributes} ~~relations~~ dependent on package-id, shipment-id, and dispense-id,

PACKAGE (package-id, batch-id, dispense-id, package-size, package-at, qr-code, status, quantity-total, pharmacy-id, dispensed-at, patient-id, remarks)

SHIPMENT (shipment-id, sender-id, receiver-id, shipment-date, status, mode)

CONTAINS (package-id, shipment-id)

DISPENSING (dispense-id, pharmacy-id, dispensed-at, patient-id, quantity, remarks)

PACKAGE U SHIPMENT U CONTAINS U DISPENSE E PACKAGE

Converting to 3NF

Removing transitive dependencies,

i) PARTY (party-id, party-name, address, phone, email)

DISTRIBUTOR (party-id, distributor-license-no)

PHARMACY (party-id, pharmacy-license-no)

REGULATOR (party-id, regulator-code)

ii) BATCH (batch_id, manufacture_date, batch_status,
drug_id, manufacturer_id)

DRUG (drug_id, drug_name, strength, dosage_form,
description)

MANUFACTURER (manufacturer_id, manufacturing_license_no)

QUALITY-TEST (test_id, batch_id, test_date, test_type,
result, status)

iii) It is already in 3NF

Converting to BCNF

For every non-trivial functional dependency,
 $X \rightarrow Y$, X must be a superkey.

The previous relations are already in BCNF.

BCNF Verification

Checking if the only determinants of relations are also candidate keys,

Relation	Candidate Keys	Determinants
PARTY	party-id	party-id
MANUFACTURER	party-id, manufacturing-license-no	both
DISTRIBUTOR	party-id, distributor-license-no	both
PHARMACY	party-id, pharmacy-license-no	both
REGULATOR	party-id, reg-code	both
DRUG	drug-id	drug-id
BATCH	batch-id	batch-id
QUALITYTEST	test-id	test-id
RECALL	recall-id	recall-id
PACKAGE	package-id	package-id
SHIPMENT	shipment-id	shipment-id
CONTAINS	package-id, shipment-id	no non-key attributes
DISPENSING	dispense-id	dispense-id

Thus, we verify all the relations are in BCNF.